La Scuola d'Italia / Dr. Huson / IB Math 11 (a)
28 October 2026

Nickname:

2.6 Early Finishers: Lines, Exponents, Radicals and Sequences

For students who finish the quiz early. A calculator is allowed.

1. [Maximum mark: 11]

   (a) Solve the equation $25^{x+1} = 125^{2x-3}$. [3]

   (b) Find the integers a and b such that $\left(2+\sqrt{3}\right)\left(a+b\sqrt{3}\right) = 12+7\sqrt{3}$. [3]

   Consider the equation $3\left(9^x\right) - 82\left(3^x\right) + 27 = 0$.

   (c) By letting $u = 3^x$, solve the equation. [5]

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2. [Maximum mark: 8]

An arithmetic sequence has first term $u_1 = 7$ and common difference 3. A geometric sequence has first term $v_1 = 2$ and common ratio 1.5.

(a) Write down an expression for u_n and an expression for v_n. [2]

(b) Find the least value of n for which $v_n > u_n$. [3]

(c) Find the least value of n for which the sum of the first n terms of the geometric sequence is greater than the sum of the first n terms of the arithmetic sequence. [3]

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2.6 Early Finishers: Lines, Exponents, Radicals and Sequences

For students who finish the quiz early. A calculator is allowed.

1. [Maximum mark: 10]

   The lines L_1 and L_2 have equations

   $$kx + 2y = 7 \qquad \text{and} \qquad 3x + (k-1)y = 5,$$

   where k is a real constant.

   (a) Find the two values of k for which L_1 and L_2 are parallel. [4]

   (b) Find the value of k for which L_1 and L_2 are perpendicular. [3]

   (c) Let $k = 1$. Find the coordinates of the point where L_1 and L_2 intersect. [3]

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2. [Maximum mark: 12]

The first three terms of a geometric sequence are $k-3$, $k+1$ and $3k-1$.

(a) Show that $k^2 - 6k + 1 = 0$. [2]

(b) Find the two possible values of k. Give your answers in the form $a + b\sqrt{2}$, where $a, b \in \mathbb{Z}$. [2]

(c) For the larger value of k, find the common ratio. Give your answer in the form $a + b\sqrt{2}$. [4]

(d) For the smaller value of k, show that the series has a sum to infinity, and find its value. [4]